Model structure comparison. MODEL equivalent at mu_x = 0, sigma_z^2 = 1; OBJECTIVE equivalent; IMPLEMENTATION differs (OLD E-step halves Y score/curvature)

item	Mikawa et al. 2024 (paper)	ExpFam Gaussian-X / Bernoulli-Y	verdict
prior of Z	N(0, sigma_z^2 I), sigma_z^2 estimated (Sect.2.2, Eq.14)	N(0, I), var_z fixed 1 + scale_Z	same at sigma_z^2=1
X | Z	N(F z + mu_x, Sigma) (Eq.1)	N(F z, Sigma) (no intercept)	same at mu_x = 0
Y | Z	Bern(sigmoid(w0 + w z_i'z_j)) (Eq.2)	same	same
pair counting	{prod_{i!=j}}^{1/2} (Eq.16)	prod over i<j	same for symmetric Y (numeric diff 0)
diagonal of Y	y_ii = 1 by convention (Sect.2.2)	unused	no effect on likelihood (i != j only)
Gaussian dispersion	Sigma diagonal, estimated (Eq.12)	Sigma diagonal, estimated	same
rotation	not stated; penalty subtracts k(k-1)/2	(Z,F)->(ZR,FR) invariant	same (numeric diff 0)
complete-data log density	ln p(Z)+ln p(X|Z)+ln p(Y|Z) (Eq.18)	same	equal at the same (Z, theta): diff 0
E-step gradient (Y part)	(1/2) sum_j (y_ij - s_ij) w z_j (Eq.23)	w sum_j (y_ij - s_ij) z_j	paper print has an extra 1/2; adopted = exact derivative of Eq.20 (FD)
E-step precision (Y part)	(1/2) sum_j s(1-s) w^2 z_j z_j' (Eq.22)	w^2 sum_j s(1-s) z_j z_j'	same as above

layer: model layer: objective layer: implementation